

Brokerage in Small-World Inventor Networks: When Do Brokers Spark Atypical Technology Combinations?

a) Importance & Relevance: regional success isn't just about more patent; it's about unusual combinations of technologies that innovate. Co-inventor networks often form small-worlds, which are highly clustered yet have short paths, making them ideal for spreading ideas locally (Watts & Strogatz, 1998). Within these networks, brokers (inventors who bridge separate communities) are key to stitching distant knowledge together. Figuring out when these brokers are most effective gives universities, labs, and regional agencies leverage to boost breakthrough innovation.

b) What We Already Know: We know that regional small-world networks are linked to innovation (Fleming et al., 2007) and that well-connected regions produce more novel (atypical) technology combinations (Abbasiharofteh et al., 2023). At the micro-level, brokers who span structural holes access diverse information and spark new ideas (Burt, 1992). This often involves a proactive tertius iungens (or 'third who joins') orientation (Obstfeld, 2005). We can measure small-worldness (Humphries & Gurney, 2008), and we know that these atypical combinations can lead to a significant long-term impact (Uzzi et al., 2013).

c) Niche & Research Question: we know small-world regions are novel, but we don't know the exact mechanism. I argue the inventor-level broker is the engine, but does that engine work better in some environments? Research question: Do regions with more brokers produce more atypical technology combinations, and crucially, is this effect amplified when the regional network is a strong small world (high clustering, short paths)? This will clarify how breakthrough ideas are fostered within a network.

d) Stakeholders & Benefits

- Universities & Public Labs: Use network analytics to find and support the specific broker scientists most likely to connect isolated departments
- Regional Innovation Agencies: Don't just fund randomly. Seed bridges (matchmaking grants, fellowships) for brokers in regions that already have a favourable small-world structure to maximise returns
- R&D Firms & Clusters: Intentionally staff teams with measured brokers to raise the probability of non-obvious tech pairings aligned with your roadmap

e) References Abbasiharofteh, M., Kogler, D. F., & Lengyel, B. (2023). Atypical combinations of technologies in regional co-inventor networks. *Research Policy*. Burt, R. S. (1992). *Structural Holes: The Social Structure of Competition*. Harvard University Press. Fleming, L., King, C., & Juda, A. I. (2007). Small Worlds and Regional Innovation. *Organization Science*, 18(6), 938–954. Humphries, M. D., & Gurney, K. (2008). Network “Small-World-Ness”: A Quantitative Method for Determining Canonical Network Equivalence. *PLOS ONE*, 3(4), e0002051. Obstfeld, D. (2005). Social Networks, the Tertius Iungens Orientation, and Involvement in Innovation. *Administrative Science Quarterly*, 50(1), 100–130. Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical Combinations and Scientific Impact. *Science*, 342(6157), 468–472. Watts, D. J., & Strogatz, S. H. (1998). Collective Dynamics of ‘Small-World’ Networks. *Nature*, 393(6684), 440–442.